GIOELE RORA

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EDUCATION

Politecnico di Torino Sept 2024 – Present
B.Sc. in Computer Engineering (Taught entirely in English)
Honors: Member, IEEE-Eta Kappa Nu (IEEE-HKN, Mu Nu Chapter) – IEEE Honor Society.
Key Coursework: Mathematical Analysis I & II, Linear Algebra & Geometry, Computer Science Principles, Physics.

Liceo Scientifico (Applied Sciences) Sept 2019 – July 2024
Diploma di Maturità
Active participant in individual and team-based Mathematics, Physics, and Informatics Olympiad competitions.

ENGINEERING EXPERIENCE & STUDENT TEAMS

Team Astra – Politecnico di Torino Sept 2024 – Oct 2026
IT Division – Spherical Lunar Rover
- Collaborating in the IT division for an autonomous spherical lunar rover powered by internal pendulum motion and robotic actuation.
- Developing, profiling, and testing embedded C/C++ firmware for DC motor velocity regulation, PWM control, and temperature sensor integration.
- Optimizing motor management driver libraries for resource-constrained microcontrollers; validated actuation systems for public exhibitions.

Team QubiTo – Politecnico di Torino Oct 2026 – Present
Software & Algorithms Division – Quantum Computing Team
- Contributing to quantum algorithm design and circuit simulation benchmarks within the university quantum computing student team.

TECHNICAL PROJECTS & HACKATHONS

EGG – Edge-AI Traffic Mesh Network | 1st Place @ HackTurin Oct 2026
Stack: Zephyr RTOS, C++, Python, Linux (Debian, Docker), Arduino UNO Q, YOLOX-nano, SQLite, RPC
- Led team to 1st place and engineered 3D hardware enclosure; co-architected a ~€100 edge-AI node to replace €15k+ proprietary traffic sensors.
- Architected dual-core firmware on Arduino UNO Q: offloaded real-time I2C sensing and traffic timers to Zephyr RTOS (C++), bridging to Debian/Docker via RPC (<7 ms latency).
- Integrated local YOLOX-nano inference on 8MP IMX219 camera for GDPR-compliant tracking; relayed pedestrian and collision alerts to a live 3D web map.

Drone Delivery Simulator – 3D Aerodynamics & Systems Game Sept 2026 – Present
Stack: Unity, C#, 3D Physics Engine, HLSL / Shader Graph, Git
- Engineering a physics flight simulator in Unity featuring real-time payload mass shifts, drag calculations, and dynamic battery drain.
- Architecting procedural mission pipelines combined with branching state logic and unlocking mechanics for vertical exploration.
- Designing a low-latency analog gamepad input framework and continuous, loading-free level deployment using modern glassmorphism shaders.

Route 2025 – Full-Stack Event Logistics Platform Jan 2025 – Present
Stack: Node.js, JavaScript (ES6+), HTML5/CSS3, Vercel, Git CI/CD | Demo: route2025.vercel.app
- Architected a centralized platform for Scout event logistics, optimizing live attendance tracking and coordination for multiple regional groups.
- Implemented modular REST endpoints and a responsive dashboard to streamline live status synchronization across 30+ concurrent participants.
- Maintained deployment pipelines via Vercel with automated continuous integration and branch protection on GitHub.

WORK & ACADEMIC EXPERIENCE

STEM Private Tutor 2021 – Present
Independent Tutor in Mathematics, Physics, and Computer Science
- Instructing high school and early college students, translating complex theoretical formulas into practical, algorithmic problem-solving methodologies.

LEADERSHIP & COMMUNITY SERVICE

Unit Leader (Capo Reparto) – AGESCI Scout Group Sept 2025 – Present
- Mentoring and directing a youth group of 30+ individuals; planning yearly educational pathways and executing national route logistics.

Volunteer Worker – Operazione Mato Grosso (OMG) 2015 – Present
- Supported infrastructure building, renovation, and operational maintenance at Rifugio delle Marmotte in high-altitude environments.

TECHNICAL SKILLS & PERSONAL INTERESTS

Languages:	C, C++, C#, Python, Java, JavaScript (ES6+), HTML5/CSS3, SQL
Systems & Embedded:	Linux (Debian, CLI), Zephyr RTOS, STM32, Arduino (UNO Q), Docker, Edge-AI, I2C/SPI, DC Motor Control
Tools & Frameworks:	Git, GitHub Actions, Unity Engine, Node.js, Vercel, SQLite
Spoken Languages:	Italian (Native), English (Professional Working Proficiency)
Interests:	Quantum Computing, Aerospace & Robotics, Embedded Systems, Game Physics, Trail Running